Tutorial 5 (to be completed in - week5)
Structured Query Language (SQL)

Tutorial Objectives:

- **Learn SQL**
  - **Write some DDL (Data Definition Language) commands.**
  - **Write some DML (Data Manipulation Language) commands.**

Tutorial Tasks:

The given database* stores data for a consulting company that tracks all charges to projects. The charges are based on the hours an employee works on each project.

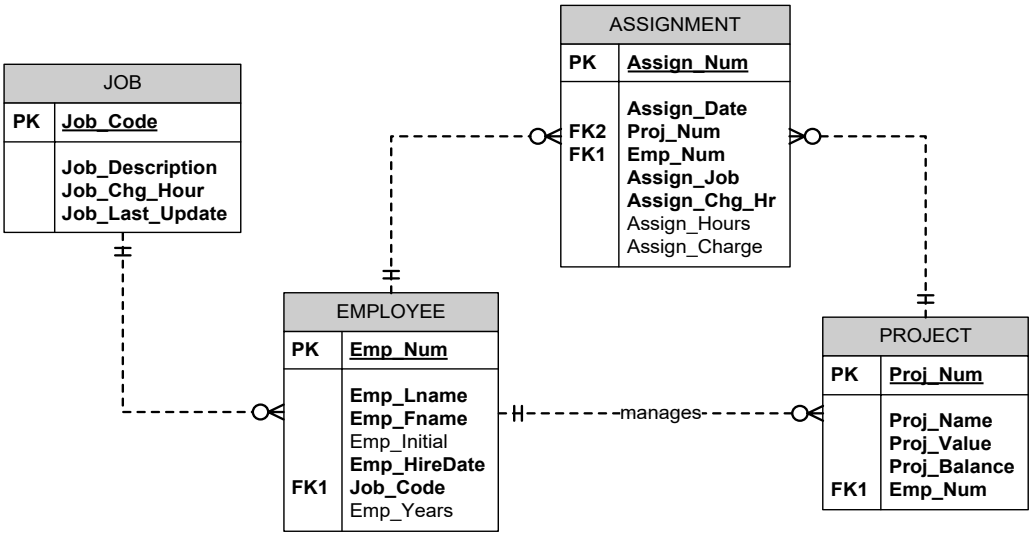


Table: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR	JOB_LAST_UPDATE
500	Programmer	35.75	20-NOV-2019
501	Systems Analyst	96.75	20-NOV-2019
502	Database Designer	125.00	24-MAR-2020
503	Electrical Engineer	84.50	20-NOV-2019
504	Mechanical Engineer	67.90	20-NOV-2019
505	Civil Engineer	55.78	20-NOV-2019
506	Clerical Support	26.87	20-NOV-2019
507	DSS Analyst	45.95	20-NOV-2019
508	Applications Designer	48.10	24-MAR-2020
509	Bio Technician	34.55	20-NOV-2019
510	General Support	18.36	20-NOV-2019

IN CONFIDENCE

* Adapted Fig P7.1 pg. 323 13ed

Table: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP	EMP_HIREDATE	JOB_CODE	EMP_YEARS
101	News	John	G	08-NOV-2000	502	22
102	Senior	David	H	12-JUL-1989	501	34
103	Arbough	June	E	01-DEC-1996	503	26
104	Ramoras	Anne	K	15-NOV-1987	501	35
105	Johnson	Alice	K	01-FEB-1993	502	30
106	Smithfield	William		22-JUN-2004	500	19
107	Alonzo	Maria	D	10-OCT-1993	500	29
108	Washington	Ralph	B	12-AUG-1991	501	32
109	Smith	Larry	W	18-JUL-1997	501	26
110	Olenko	Gerald	A	11-DEC-1995	505	27
111	Wabash	Geoff	B	04-APR-1991	506	32
112	Smithson	Darlene	M	23-OCT-1994	507	28
113	Joenbrood	Delbert	K	15-NOV-1996	508	26
114	Jones	Annelise		10-AUG-1993	508	30
115	Bawangi	Travis	B	25-JAN-1992	501	31
116	Pratt	Gerald	L	05-MAR-1997	510	26
117	Williamson	Angie	H	19-JUN-1996	509	27
118	Frommer	James	J	04-JAN-2005	510	18

Table: PROJECT

PROJ_N	PROJ_NAME	PROJ_VALUE	PROJ_BALANCE	EMP_NUM
15	Evergreen	1453500.00	1002350.00	103
18	Amber Wave	3500500.00	2110346.00	108
22	Rolling Tide	805000.00	500345.20	102
25	Starflight	2650500.00	2309880.00	107

Table: ASSIGNMENT

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_JO	ASSIGN_CHG_HR	ASSIGN_HOURS	ASSIGN_CHARGE
1001	22-MAR-2020	18	103	503	84.50	3.5	295.75
1002	22-MAR-2020	22	117	509	34.55	4.2	145.11
1003	22-MAR-2020	18	117	509	34.55	2.0	69.10
1004	22-MAR-2020	18	103	503	84.50	5.9	498.55
1005	22-MAR-2020	25	108	501	96.75	2.2	212.85
1006	22-MAR-2020	22	104	501	96.75	4.2	406.35
1007	22-MAR-2020	25	113	508	50.75	3.8	192.85
1008	22-MAR-2020	18	103	503	84.50	.9	76.05
1009	23-MAR-2020	15	115	501	96.75	5.6	541.80
1010	23-MAR-2020	15	117	509	34.55	2.4	82.92
1011	23-MAR-2020	25	105	502	105.00	4.3	451.50
1012	23-MAR-2020	18	108	501	96.75	3.4	328.95
1013	23-MAR-2020	25	115	501	96.75	2.0	193.50
1014	23-MAR-2020	22	104	501	96.75	2.8	270.90
1015	23-MAR-2020	15	103	503	84.50	6.1	515.45
1016	23-MAR-2020	22	105	502	105.00	4.7	493.50
1017	23-MAR-2020	18	117	509	34.55	3.8	131.29
1018	23-MAR-2020	25	117	509	34.55	2.2	76.01
1019	24-MAR-2020	25	104	501	110.50	4.9	541.45
1020	24-MAR-2020	15	101	502	125.00	3.1	387.50
1021	24-MAR-2020	22	108	501	110.50	2.7	298.35
1022	24-MAR-2020	22	115	501	110.50	4.9	541.45
1023	24-MAR-2020	22	105	502	125.00	3.5	437.50
1024	24-MAR-2020	15	103	503	84.50	3.3	278.85
1025	24-MAR-2020	18	117	509	34.55	4.2	145.11

a. Run the script Ch07_ConstructCo_ORA.sql provided on the course Stream site to create and populate the four tables.

b. Open a new text file (e.g., Notepad) to write SQL commands for the following two sets of questions. Alternatively, start writing the code in SQL Developer and later save your script in a file.

DDL / DML / TCL Questions:

1. Write the SQL code to create the table structure for a table named EMP_1. This table is a subset of the EMPLOYEE table. The basic EMP_1 table structure is summarised in Table 1. (Note that the JOB_CODE is the FK to JOB.)

Table 1: The EMP_1 Table Structure

ATTRIBUTE (FIELD) NAME	DATA DECLARATION
EMP_NUM	CHAR (3)
EMP_LNAME	VARCHAR2(15)
EMP_FNAME	VARCHAR2(15)
EMP_INITIAL	CHAR (1)
EMP_HIREDATE	DATE
JOB_CODE	CHAR (3)

2. Write the SQL code to insert values (this is DML but needed to populate here) for the nine rows in EMP_1 as shown below.

Table 2: The Contents of the EMP_1 Table

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-00	502
102	Senior	David	H	12-Jul-89	501
103	Arbough	June	E	01-Dec-96	500
104	Ramoras	Anne	K	15-Nov-87	501
105	Johnson	Alice	K	01-Feb-93	502
106	Smithfield	William		22-Jun-04	500
107	Alonzo	Maria	D	10-Oct-93	500
108	Washington	Ralph	B	22-Aug-91	501
109	Smith	Larry	W	18-Jul-97	501

3. Write the SQL code that will save the changes made to the EMP_1 table.

4. Write the SQL code to create a copy of EMP_1, naming the copy EMP_2. Next, write the SQL code that will add the attributes EMP_PCT and PROJ_NUM to its structure. The EMP_PCT is the bonus percentage to be paid to each employee. The characteristics for the new attributes are:
EMP_PCT Number (4, 2) PROJ_NUM Char (3)

Write the SQL code (update is DML) to enter an EMP_PCT value of 3.85 for the person whose employee number (EMP_NUM) is 103. Next, enter the remaining EMP_PCT values shown in Table 3.

Table 3: The Contents of the EMP_2 Table after the modifications

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE	EMP_PCT	PROJ_NUM
101	News	John	G	08-Nov-00	502	5.00	
102	Senior	David	H	12-Jul-89	501	8.00	
103	Arbough	June	E	01-Dec-96	500	3.85	
104	Ramoras	Anne	K	15-Nov-87	501	10.00	
105	Johnson	Alice	K	01-Feb-93	502	5.00	
106	Smithfield	William		22-Jun-04	500	6.20	
107	Alonzo	Maria	D	10-Oct-93	500	5.15	
108	Washington	Ralph	B	22-Aug-91	501	10.00	
109	Smith	Larry	W	18-Jul-97	501	2.00	

5. Write two SQL commands to:

   a. Create a temporary table named TEMP_1 whose structure is composed of the EMP_2 attributes EMP_NUM and EMP_PCT.
   b. Copy the matching EMP_2 values into the TEMP_1 table.

6. Write a SQL command that will delete the newly created TEMP_1 table from the database.

DML Questions:

1. Write the SQL code required to list the employee number, last name, first name, and middle initial of all employees whose last names start with *Smith*. In other words, the rows for both Smith and Smithfield should be included in the listing. Sort the results by employee number.

2. Using the given EMPLOYEE, JOB, and PROJECT tables, write the SQL code that will join the EMPLOYEE and PROJECT tables using EMP_NUM as the common attribute. Display the attributes shown in the results presented in Figure P7.2, sorted by project value.

Table 4: The Query Results for Question 2

FIGURE P7.2 THE QUERY RESULTS FOR PROBLEM 2

PROJ_NAME	PROJ_VALUE	PROJ_BALANCE	EMP_LNAME	EMP_FNAME	EMP_INITIAL	JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
Rolling Tide	805000.00	500345.20	Senior	David	H	501	Systems Analyst	96.75
Evergreen	1453500.00	1002350.00	Arbough	June	E	500	Programmer	35.75
Starflight	2650500.00	2309880.00	Alonzo	Maria	D	500	Programmer	35.75
Amber Wave	3500500.00	2110346.00	Washington	Ralph	B	501	Systems Analyst	96.75

3. Write the SQL code that will produce the same information that was shown in Q2 but sorted by the employee's last name.

4. Write the SQL code to delete the rows for employees William Smithfield and Darlene Smithson. (Use a **single** statement only.)

   Now, write the SQL code that will restore the data to its original status.

5. Write the SQL code that will list only the distinct project numbers found in the Assignment table, sorted by project number.

6. Write the SQL code to validate the ASSIGN_CHARGE values in the ASSIGNMENT table. Your query should retrieve the assignment number, employee number, project number, the stored assignment charge (ASSIGN_CHARGE), and the calculated assignment charge (calculated by multiplying ASSIGN_CHG_HR by ASSIGN_HOURS). Sort the results by the assignment number.

7. Using the data in the ASSIGNMENT table, write the SQL code that will yield the total number of hours worked for each employee and the total charges stemming from those hours worked, sorted by employee number. The results of running that query are shown below.

Table 5: Total Hours and Charges by Employee

EMP_NUM	EMP_LNAME	SumOfASSIGN_HOURS	SumOfASSIGN_CHARGE
101	News	3.1	387.50
103	Arbough	19.7	1664.65
104	Ramoras	11.9	1218.70
105	Johnson	12.5	1382.50
108	Washington	8.3	840.15
113	Joenbrood	3.8	192.85
115	Bawangi	12.5	1276.75
117	Williamson	18.8	649.54

8. Write a query to produce the total number of hours and charges for each of the projects represented in the ASSIGNMENT table. The output is shown below.

Table 6: Total Hours and Charges by Project

PROJ_NUM	SumOfASSIGN_HOURS	SumOfASSIGN_CHARGE
15	20.5	1806.52
18	23.7	1544.80
22	27.0	2593.16
25	19.4	1668.16

9. Write the SQL code to generate the total hours worked and the total charges made by all employees. The results are shown in Table 7.

Table 7: Total Hours and Charges, All Employees

SumOfSumOfASSIGN_HOURS	SumOfSumOfASSIGN_CHARGE
90.6	7612.64

10. Write the SQL code to get a list of employees (employee name, employee code) along with the projects (project number, project name) they have been assigned to. The results should be sorted by the project number and should not display any duplicates.

Notes: Listed below are few formatting suggestions for the tutorials that you can use with your query results. Refer to SQL functions in Chapter 8 of your textbook.

- Modify some of the queries that list employee names as a single column instead of separate columns for First name, Last name etc.
- Rename some of the default column names to make them more readable (e.g., use HireDate or Hire Date instead of Emp_Hiredate).
- Include formatting information for columns with currency values (e.g., Proj_Balance value $5,00,345.20 instead of 500345.2)